

# **.NET Interview Guide (3 Years Experience)**

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# 1. Introduction to .NET

## What is .NET?

.NET is a free, open-source, cross-platform development framework developed by Microsoft.

## Components of .NET

- CLR (Common Language Runtime)
- CTS (Common Type System)
- CLS (Common Language Specification)
- BCL (Base Class Library)
- ASP.NET Core
- Entity Framework Core

## .NET Versions

- .NET Framework
- .NET Core
- .NET 5+
- .NET 6 / 7 / 8 (LTS and STS)

## Interview Questions

1. Difference between .NET Framework and .NET Core?
  2. What is CLR?
  3. What is managed code?
  4. What is CTS and CLS?
  5. Explain JIT Compiler.
- 

# 2. C# Fundamentals

## Data Types

- Value Types
- Reference Types

## Variables

- local
- static
- const
- readonly

## Access Modifiers

- public
- private
- protected
- internal
- protected internal

## Operators

- Arithmetic
- Logical
- Relational
- Null-coalescing

## Control Statements

- if/else
- switch
- loops

## String Handling

- String vs StringBuilder

## Interview Questions

1. Difference between value type and reference type?
  2. Difference between const and readonly?
  3. Why String is immutable?
  4. Difference between == and Equals?
  5. Difference between ref and out?
- 

# 3. Object-Oriented Programming (OOP)

## Concepts

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

## Types of Polymorphism

- Compile-time
- Runtime

## Keywords

- virtual
- override
- abstract
- sealed

## Interview Questions

1. What is abstraction?
  2. Difference between abstraction and encapsulation?
  3. Can we create object for abstract class?
  4. Difference between interface and abstract class?
  5. What is method overriding?
- 

# 4. Advanced C# Concepts

## Topics

- Extension Methods
- Anonymous Types
- Dynamic Keyword
- Nullable Types
- Indexers
- Properties
- Records
- Tuples
- Pattern Matching

## Interview Questions

1. What are extension methods?
  2. Difference between var and dynamic?
  3. What are records in C#?
  4. Explain nullable reference types.
-

## 5. Collections & Generics

### Collections

- Array
- ArrayList
- List<T>
- Dictionary<TKey, TValue>
- HashSet<T>
- Queue<T>
- Stack<T>

### Generics

Benefits:

- Type safety
- Performance
- Reusability

### Interview Questions

1. Difference between Array and ArrayList?
  2. Difference between List and Dictionary?
  3. What is HashSet?
  4. Why generics are used?
  5. Difference between IEnumerable and IQueryable?
- 

## 6. Exception Handling

### Keywords

- try
- catch
- finally
- throw
- custom exceptions

### Best Practices

- Avoid empty catch blocks
- Use finally for cleanup
- Create meaningful exception messages

## Interview Questions

1. Difference between throw and throw ex?
  2. What is custom exception?
  3. Can finally block execute without catch?
- 

# 7. Delegates, Events & Lambda Expressions

## Delegates

Type-safe function pointers.

## Built-in Delegates

- Action
- Func
- Predicate

## Events

Publisher-subscriber model.

## Lambda Expressions

Short syntax for anonymous methods.

## Interview Questions

1. What is delegate?
  2. Difference between delegate and event?
  3. What is multicast delegate?
  4. Explain lambda expression.
- 

# 8. LINQ

## LINQ Types

- LINQ to Objects
- LINQ to SQL
- LINQ to Entities

## Common Methods

- Select
- Where
- OrderBy
- GroupBy
- Join
- Any
- All
- FirstOrDefault

## Deferred Execution

Query executes only when iterated.

## Interview Questions

1. What is LINQ?
  2. Difference between First and FirstOrDefault?
  3. Difference between IEnumerable and IQueryable?
  4. What is deferred execution?
- 

# 9. Asynchronous Programming

## Keywords

- async
- await
- Task
- Task<T>

## Concepts

- Synchronous vs Asynchronous
- Parallel Programming

## Interview Questions

1. What is async-await?
  2. Difference between Task and Thread?
  3. What is deadlock?
  4. Difference between async and parallel programming?
-

# 10. Memory Management & Garbage Collection

## Garbage Collection Generations

- Gen 0
- Gen 1
- Gen 2

## Managed Heap

Memory managed by CLR.

## IDisposable

Used for unmanaged resources.

## Interview Questions

1. What is garbage collector?
  2. Difference between Dispose and Finalize?
  3. What is memory leak?
- 

# 11. .NET Runtime & CLR

## CLR Responsibilities

- Memory Management
- Exception Handling
- Thread Management
- Security
- JIT Compilation

## Interview Questions

1. What is CLR?
  2. What is MSIL?
  3. Explain JIT compilation.
-



## 12. ASP.NET Core

### Features

- Cross-platform
- High performance
- Built-in Dependency Injection
- Middleware pipeline

### Project Structure

- Program.cs
- appsettings.json
- Controllers
- Services

### Interview Questions

1. What is ASP.NET Core?
  2. Difference between ASP.NET MVC and ASP.NET Core?
  3. Explain middleware.
  4. What is Kestrel?
- 

## 13. Web API

### REST Principles

- Stateless
- Client-server
- Cacheable

### HTTP Methods

- GET
- POST
- PUT
- DELETE
- PATCH

### Status Codes

- 200 OK
- 201 Created

- 400 Bad Request
- 401 Unauthorized
- 404 Not Found
- 500 Internal Server Error

## Interview Questions

1. What is REST API?
  2. Difference between PUT and PATCH?
  3. What is content negotiation?
  4. How to version APIs?
- 

# 14. Entity Framework Core

## Approaches

- Code First
- Database First

## Important Concepts

- DbContext
- DbSet
- Migrations
- Tracking
- Lazy Loading
- Eager Loading

## Interview Questions

1. What is EF Core?
  2. Difference between FirstOrDefault and SingleOrDefault?
  3. What is migration?
  4. Explain lazy loading.
  5. What is change tracking?
- 

# 15. Authentication & Authorization

## Authentication Methods

- JWT
- Cookies

- OAuth
- Identity

## Authorization

Role-based and policy-based.

## JWT Structure

- Header
- Payload
- Signature

## Interview Questions

1. Difference between authentication and authorization?
  2. What is JWT?
  3. How JWT works?
  4. What is OAuth?
- 

# 16. Dependency Injection

## Service Lifetimes

- Singleton
- Scoped
- Transient

## Benefits

- Loose coupling
- Testability
- Maintainability

## Interview Questions

1. What is Dependency Injection?
  2. Difference between Singleton and Scoped?
  3. What problems DI solves?
-

# 17. Middleware

## Built-in Middleware

- Authentication
- Authorization
- Routing
- Exception Handling

## Custom Middleware

```
public class MyMiddleware
{
    private readonly RequestDelegate _next;

    public MyMiddleware(RequestDelegate next)
    {
        _next = next;
    }

    public async Task Invoke(HttpContext context)
    {
        await _next(context);
    }
}
```

## Interview Questions

1. What is middleware?
  2. How middleware pipeline works?
  3. Difference between middleware and filters?
- 

# 18. Logging & Monitoring

## Logging Providers

- Console
- Debug
- Serilog
- NLog

## Monitoring Tools

- Application Insights
- Prometheus
- Grafana

## Interview Questions

1. What is structured logging?
  2. Difference between logging and monitoring?
- 

# 19. Microservices

## Concepts

- API Gateway
- Service Discovery
- Event-driven architecture
- Message Broker

## Tools

- RabbitMQ
- Kafka
- Docker
- Kubernetes

## Interview Questions

1. What are microservices?
  2. Difference between monolith and microservices?
  3. What is API Gateway?
- 

# 20. Design Patterns

## Creational

- Singleton
- Factory

## Structural

- Adapter
- Decorator

## Behavioral

- Observer
- Strategy

## Interview Questions

1. Explain Singleton pattern.
  2. What is Factory pattern?
  3. Where have you used design patterns?
- 

# 21. SOLID Principles

## Principles

- S - Single Responsibility
- O - Open Closed
- L - Liskov Substitution
- I - Interface Segregation
- D - Dependency Inversion

## Interview Questions

1. Explain SOLID principles.
  2. Real-time example of SRP?
  3. What is Dependency Inversion?
- 

# 22. Unit Testing

## Frameworks

- xUnit
- NUnit
- MSTest

## Mocking

- Moq

## Interview Questions

1. What is unit testing?
  2. Difference between unit and integration testing?
  3. What is mocking?
- 

## 23. Caching

### Types

- In-memory Cache
- Distributed Cache
- Redis

## Interview Questions

1. What is caching?
  2. Difference between Redis and MemoryCache?
- 

## 24. SignalR

### Features

- Real-time communication
- WebSockets

### Use Cases

- Chat apps
- Notifications
- Live dashboards

## Interview Questions

1. What is SignalR?
  2. How SignalR works?
-

## 25. Docker & Kubernetes Basics

### Docker

- Containerization
- Dockerfile
- Images
- Containers

### Kubernetes

- Pods
- Deployments
- Services

### Interview Questions

1. What is Docker?
  2. Difference between VM and Container?
  3. What is Kubernetes?
- 

## 26. Azure Basics

### Azure Services

- App Service
- Azure Functions
- Azure SQL
- Blob Storage
- Azure DevOps

### Interview Questions

1. What is Azure App Service?
  2. What is Azure Function?
  3. Explain CI/CD.
-



## 27. Multithreading

### Concepts

- Thread
- Task
- Parallel
- Lock
- Semaphore

### Interview Questions

1. What is multithreading?
  2. Difference between Thread and Task?
  3. What is race condition?
  4. What is lock keyword?
- 

## 28. Performance Optimization

### Techniques

- Caching
- Async Programming
- Pagination
- Compression
- Database indexing

### Interview Questions

1. How to improve API performance?
  2. What is pagination?
  3. What causes slow EF queries?
- 

## 29. Security Best Practices

### Topics

- HTTPS
- JWT Security
- SQL Injection Prevention
- XSS Prevention

- CSRF

## Interview Questions

1. What is SQL Injection?
  2. How to secure APIs?
  3. What is CORS?
- 

## 30. REST & HTTP Concepts

### HTTP Headers

- Authorization
- Content-Type
- Accept

### Stateless Communication

Each request is independent.

### Interview Questions

1. What is idempotent?
  2. Difference between HTTP and HTTPS?
  3. Explain REST constraints.
- 

## 31. System Design Basics

### Topics

- Scalability
- Load Balancer
- Database Sharding
- Caching
- Message Queues

### Interview Questions

1. Design URL shortener.
2. Design notification service.
3. How to scale APIs?

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## 32. Common Interview Questions

### Technical

1. Explain your current project architecture.
  2. Explain your role in the project.
  3. How do you handle exceptions globally?
  4. How do you optimize APIs?
  5. Explain your deployment process.
  6. Explain repository pattern.
  7. Explain Unit of Work pattern.
  8. What challenges have you faced?
  9. Explain async calls in your project.
  10. Explain JWT implementation.
- 

## 33. Coding Problems

### Reverse String

```
string input = "hello";  
string reversed = new string(input.Reverse().ToArray());
```

### Palindrome

```
bool IsPalindrome(string s)  
{  
    return s == new string(s.Reverse().ToArray());  
}
```

### Fibonacci

```
int Fib(int n)  
{  
    if (n <= 1) return n;  
    return Fib(n - 1) + Fib(n - 2);  
}
```

## LINQ GroupBy

```
var result = employees.GroupBy(x => x.Department);
```

## 34. HR & Scenario-Based Questions

### HR Questions

1. Tell me about yourself.
2. Why do you want to switch?
3. What are your strengths?
4. What are your weaknesses?
5. Where do you see yourself in 5 years?

### Scenario Questions

1. Production issue handling?
2. Conflict with teammate?
3. Missed deadline handling?
4. Difficult bug debugging approach?

## 35. Quick Revision Notes

### Important Differences

#### String vs StringBuilder

| String                   | StringBuilder |
|--------------------------|---------------|
| Immutable                | Mutable       |
| Slower for modifications | Faster        |

#### Abstract Class vs Interface

| Abstract Class          | Interface            |
|-------------------------|----------------------|
| Can have implementation | Mostly contract      |
| Single inheritance      | Multiple inheritance |

## IEnumerable vs IQueryable

| IEnumerable     | IQueryable       |
|-----------------|------------------|
| In-memory       | Database query   |
| LINQ to Objects | LINQ to Entities |

## Task vs Thread

| Task                   | Thread            |
|------------------------|-------------------|
| Lightweight            | Heavyweight       |
| Managed by thread pool | Manual management |

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# Important Topics for 3 Years Experience

## Must Prepare Thoroughly

- OOPs
- SOLID Principles
- ASP.NET Core
- Web API
- Entity Framework Core
- LINQ
- Async/Await
- Dependency Injection
- JWT Authentication
- Middleware
- Exception Handling
- SQL Basics
- Design Patterns
- Azure Basics
- Docker Basics

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# SQL Topics to Prepare

## Queries

- Joins
- Group By
- Having

- Subqueries
- Stored Procedures
- Indexes
- Transactions

## SQL Interview Questions

1. Difference between DELETE, TRUNCATE, DROP?
  2. Clustered vs Non-clustered Index?
  3. What is normalization?
  4. ACID properties?
  5. What is deadlock?
- 

## Final Interview Tips

1. Be strong in fundamentals.
  2. Explain with real-time examples.
  3. Prepare project architecture clearly.
  4. Practice coding daily.
  5. Revise HTTP status codes.
  6. Understand API lifecycle.
  7. Know deployment basics.
  8. Practice SQL queries.
  9. Learn debugging techniques.
  10. Prepare scenario-based answers.
- 

## Recommended Learning Resources

### Official Documentation

- Microsoft Learn
- ASP.NET Core Docs
- Entity Framework Docs

### Practice Platforms

- LeetCode
- HackerRank
- CodeWars

## YouTube Channels

- DotNetMastery
  - Raw Coding
  - Nick Chapsas
- 

## Advanced Topics (Optional but Valuable)

### Advanced ASP.NET Core

- Minimal APIs
- gRPC
- API Rate Limiting
- Background Services
- Hosted Services

### Cloud & DevOps

- Azure DevOps Pipelines
- GitHub Actions
- Terraform Basics

### Architecture

- Clean Architecture
  - CQRS
  - MediatR
  - Event Sourcing
- 

## Clean Architecture Layers

1. Domain Layer
2. Application Layer
3. Infrastructure Layer
4. Presentation Layer

### Benefits

- Separation of concerns
  - Better testability
  - Scalability
-

## Repository Pattern Example

```
public interface IUserRepository
{
    Task<User> GetByIdAsync(int id);
}

public class UserRepository : IUserRepository
{
    private readonly AppDbContext _context;

    public UserRepository(AppDbContext context)
    {
        _context = context;
    }

    public async Task<User> GetByIdAsync(int id)
    {
        return await _context.Users.FindAsync(id);
    }
}
```

---

## Middleware Registration

```
app.UseAuthentication();
app.UseAuthorization();
app.MapControllers();
```

---

## JWT Authentication Flow

1. User logs in.
  2. Server validates credentials.
  3. JWT token generated.
  4. Client stores token.
  5. Token sent in Authorization header.
  6. Server validates token.
-



# Common API Best Practices

1. Use proper status codes.
  2. Use DTOs.
  3. Validate input.
  4. Global exception handling.
  5. Use async methods.
  6. Secure endpoints.
  7. Version APIs.
  8. Add logging.
  9. Use pagination.
  10. Add Swagger documentation.
- 

## Swagger Setup

```
builder.Services.AddSwaggerGen();  
  
app.UseSwagger();  
app.UseSwaggerUI();
```

---

## Global Exception Handling Middleware

```
app.UseExceptionHandler("/error");
```

---

## Conclusion

This guide covers the most important .NET topics commonly asked in interviews for candidates with around 3 years of experience.

Focus more on:

- Real-time project explanations
- API design
- EF Core
- Async programming
- Design patterns
- SQL

- Problem-solving

Consistent revision and mock interviews will significantly improve your confidence.